Forest Ecology and Management 257 (2009) 2098–2107

Contents lists available at ScienceDirect

Forest Ecology and Management

journal homepage: www.elsevier.com/locate/foreco

The impact of fire and density-dependent mortality on the spatial patterns of a pine forest in the Hulun Buir sandland, Inner Mongolia, China

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A R T I C L E I N F O

Article history:
Received 16 October 2008
Received in revised form 7 February 2009
Accepted 13 February 2009

Keywords:
Disturbance
Hulun Buir sandland
Mongolian pine
Pair correlation function
Point pattern analysis
Self-thinning
Spatial pattern
Stand
Surface fire

A B S T R A C T

Wildland fire, especially surface fire, is one of the major disturbances in forest ecosystems such as the Mongolian pine forest of the Hulun Buir sandland. However, little is known about the impact of fire on the spatial patterns at the stand level and its consequences for successional dynamics. To fill this gap we mapped three plots of Mongolian pine forest comprising sites of the same age that were affected by fire in 2006 and 1994, and a site without fire. We explored the stand structure using diameter at breast height (DBH) distributions and basal area, and used point pattern analysis to quantify the observed spatial patterns. Null models included homogeneous Poisson and Thomas cluster processes for univariate analyses, toroidal shift to test for independence, and random labelling for analyzing mortality. Large stems showed at all three stands a tendency to regularity, whereas small stems were mostly clustered. However, small stems did not become more regular at plots with earlier fire, and only the "overstocked" 2006 pre-fire stand showed negative association between large and small trees. Mortality was non-random, showing clustering of dead (or surviving) trees and clear density dependence where stems with more neighbors had a higher risk to die. Fire killed mostly smaller trees and after prolonged fire-free periods the stand approached the pre-fire patterns structurally and spatially. Our study showed that surface fire causes strong thinning in later succession stages and pushes the forest into maturity. It may also enhance resistance of our forest to more severe fire under the relatively harsh environmental conditions. A novel test statistic for random labeling allowed directly testing for density-dependent mortality and could be widely applied in point pattern analysis.

1. Introduction

Disturbances such as fire, windstorms, floods, and human activities have been widely recognized as important driving forces in many ecosystems (Pickett and White, 1985; Mackey and Currie, 2000; White and Jentsch, 2001; Sutherland, 2007). Fire is commonly considered as one of the main sources of terrestrial disturbances and an integral factor of the dynamics of forests (Spurr, 1964; Wright and Bailey, 1982; Pyne, 1984; Barnes et al., 1998) and many other ecosystems (Bond and van Wilgen, 1996; Whelan, 1996). Plants in fire-prone ecosystems exhibit a wide range of versatile responses to fire (Gill, 1975; Keeley, 1991; Carrington, 1999; Lesica, 1999; Liu and Menges, 2005; Govender et al., 2006).

Low-intensity (surface) fires may cause differential thinning of trees in relation to size or species (Fulé and Covington, 1998). Smaller stems, although projecting out of the flame, are less likely to survive a surface fire due to igniting or heating than larger stems. The risk of mortality may also vary with local fluctuations in fire severity due to differences in local fuel quantity and quality. Fuel accumulation under aggregated trees, especially of pines which produce resinous fuels, may result in more intense fire behavior (Fulé and Covington, 1998; Rebertus et al., 1989). The interplay of these mechanisms may cause low-intensity fires to remove smaller and more fire-susceptible trees (Fulé and Laughlin, 2007) and thin clumpy patches of smaller trees to a more random or uniform spatial distribution of isolated clumps (Rebertus et al., 1989). As a result, surface fire tends to exclude fire-susceptible species in a stand, and can suppress or exclude fire-sensitive competitors of individual plants (Agee, 1993; Bond and van Wilgen, 1996; Quintana-Ascencio and Morales-Hernández, 1997; Peterson and Reich, 2001; Fulé and Laughlin, 2007).

At the stand scale fire creates a heterogeneous pattern (Pickett and White, 1985; Lindbladh et al., 2000; Peterson and Reich, 2001), and at the landscape scale a mosaics of burned vs. unburned patches (Romme, 1982; Foster et al., 1998; Syphard et al., 2007).

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doi:10.1016/j.foreco.2009.02.019

Thus, the stand structure and spatial distributions of the stems may experience dramatic changes after fire and the spatial patterns and spatial variation in stand structure crated by fire can have important effects on the subsequent succession of forest ecosystems at the stand scale. Fire is therefore not just a tragic event for the forest but has profound implications for its long-term dynamics. Note that over the last several decades a great deal of labor and money has been invested to extinguish forest fires in northeast China (Bi, 1993).

One approach for better understanding how fire shapes the dynamics of forest ecosystems is to analyse the response of spatial pattern to fire, and to explore how the spatial patterns change over time (e.g., Kenkel, 1988; Duncan, 1991; Fulé and Covington, 1998; Kenkel et al., 1997; He and Duncan, 2000; Stoyan and Pettinen, 2000; Debski et al., 2002; Getzin et al., 2006). This requires methods of spatial pattern analysis to describe the small-scale spatial correlation structure of the spatial pattern of stems in a forest. Spatial point pattern analysis using second-order statistics such as Ripley's *K*-function (Ripley, 1976, 1977) or the pair correlation function (Stoyan and Stoyan, 1994; Illian et al., 2008) is powerful method for this purpose (Andersen, 1992; Dale, 1999; Getzin et al., 2006; Montes and Cañellas, 2007), and can reveal non-random spatial structures in tree mortality (Kenkel, 1988).

In this study, we use spatial pattern analysis to infer, from the fine-scale spatial distributions of trees, how the dominant tree species in a forest compete and partition space, and how their spatial interactions change in response to fire. More specifically, we analyse a unique data set of Mongolian pine (*Pinus sylvestris* L. var. *mongolica* Litv.) dominated forest sites in the Hulun Buir sandland of northeast China. The forest was subjected to severe disturbances by wildfires and anthropogenic activities. However, this area was protected in the 1950s from commercial timber extraction and the present forest originated naturally from seeds (Zhao and Li, 1963). One of the stands has remained fire free for 38 years (the fire-excluded stand), and the two others were affected by surface fire events on April 16–22, 1994 (the 1994-burned stand) and May 16–18, 2006 (the 2006-burned stand). These surface fire events provide a unique opportunity for exploring spatial and structural changes in response to fire and allow for an assessment of the ecological consequences of the current (fire suppression) and alternative management practice in this forest. For example, moderate harvest of the smaller trees for fuel purposes in rural areas with severe shortages of timber and fuelwood may also reduce the risk of catastrophic fires in overstocked forest stands. Before analyzing spatial patterns, we compare the structural characteristics of the stems and the intensities of the surface fires among the three sites with one-way variance analysis (ANOVA).

Our analyses are guided by the following hypotheses. First, we expect stark differences in stand structure characteristics with time since fire (analysis 1). Second, stems in a stand that has been fire free for longer periods should show more regular spatial patterns, especially in larger trees, due to self-thinning (Kenkel, 1988) (analysis 2). Third, we expect negative spatial associations between small and large trees to arise due to competition (analysis 3). Fourth, the spatial patterns of surviving trees should be more regular than the pre-mortality pattern, either due to competitive self-thinning or due to fire which thins clumps of fire-susceptible (smaller) trees (analysis 4). We expect that dead trees are more clumped than expected under random mortality (e.g., Kenkel, 1988; Rebertus et al., 1989; Duncan, 1991; Newton and Jolliffe, 1998; Mast and Veblen, 1999; He and Duncan, 2000; Park, 2003). Additionally, we expect mortality to be spatially correlated in a density-dependent way where stems located in areas of higher stem density have a higher likelihood of mortality.

2. Methods

2.1. Study sites and data collection

Our study site is located at Honghuaerji of the Hulun Buir sandland, Inner Mongolia, China (48°15′N, 120°0.5′E), which is located in a vegetation gradient of a forest-to-steppe ecotone between the Da xing'anling mountains and the Mongolian plateau. Its physiognomy is characterized by hills of the Quaternary's sand sediments covering dozens to hundreds of meters (Ci, 2005). Climate is semiarid with cool and short summers, cold and long winters, a mean annual temperature of 1.5 °C below zero, and a mean annual precipitation of 344 mm supplied mainly in July and August (Zhao and Li, 1963). Elevation of the study sites range between 700 and 1100 m above sea level. The dominant species in our study area is Mongolian pine, one of three varieties of Scots pine (*P. sylvestris* L.) (Wu, 1956; Farjon, 1998). The upper canopy of the forest is almost exclusively made up of pine, but in the lower canopy and understory occur birch (*Betula platyphylla* Suk.), aspen (*Populus davidiana* Dode), and other several woody species [e.g., *Malus baccata* (L.) Borkh., *Armeniaca sibirica* (L.) Lamarck, *Crataegus dahurica* Koehne ex C.K. Schneider].

Data were collected on three 1 ha plots located in the broad plain valley and on moderate slopes with elevations ranging from 848 to 931 m above sea level. Environmental factors within the three sites showed no apparent heterogeneity. All stumps, fallen logs, standing dead trees (referring to fire-induced dead trees in the 2006-burned stand), and living trees were mapped in July and August of 2006 and 2007 (Fig. 1). Diameter at breast height (DBH) of each fallen log, standing dead tree, and living tree were measured at 1.3 m from the base (referring to fallen logs) or above ground, respectively, and diameters of stumps were also measured at the base. All char heights of stem that we could identify were measured. All individuals were identified to species level following the *Flora Republicae Popularis Sinicae*. We focus only on the dominant species (pines and birches). For the purpose of our analyses we divided the data into small trees (DBH $<$ 15 cm) and large trees (DBH $\geq$ 15 cm).

2.2. Surface fire characteristics and stand structure (analysis 1)

Average bole char height of stems in each plot was used as a surrogate for surface fire intensity (Regelbrugge and Conard, 1993; Waldrop and Brose, 1999; Menges and Deyrup, 2001). Mean DBHs of the stem and its basal areas at 1.3 m aboveground were used to characterize the stand structure. This information is used to assess the "temporal memory" of forest with respect to the fire disturbance, i.e., to find out if the upperstory structures of the stands had already converged to undisturbed stands of similar origins.

Conversion of diameter at the base data to DBH was carried out if DBH was not measurable for stumps (HFA, 2005). Nonparametric one-way analysis of variance (ANOVA) and Duncan's multiple range tests were conducted to detect between plots differences in char heights (an indicator of surface fire intensity) and mean DBH for size classes constructed with 5-cm class intervals. Differences in stand structure were also explored based on the size class distribution of trees. All of these analyses were carried out in SAS 9.0 (SAS Institute, 2002).

2.3. Spatial pattern analysis

We used recent methods of spatial point pattern analysis (Stoyan and Stoyan, 1994; Diggle, 2003; Illian et al., 2008) to analyze the spatial patterns found at our three study plots. We used univariate and bivariate pair correlation functions $g(r)$ and

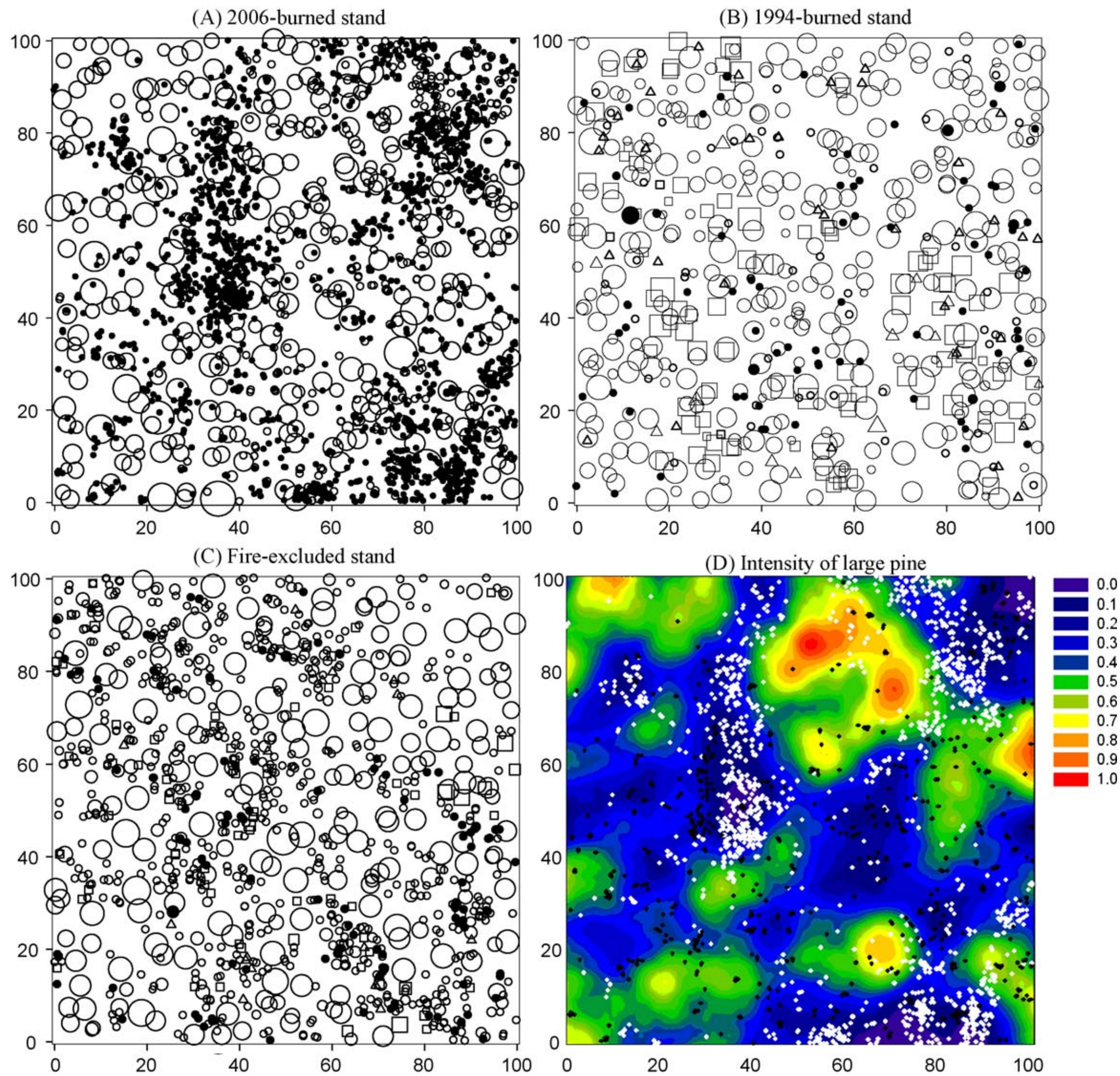

Fig. 1. Maps of the 2006-burned stand (A), the 1994-burned stand (B), and the fire-excluded stand (C) plots. Symbols are stumps (□), fallen logs (△), standing dead trees (fire-induced standing dead trees in A) (●), and living trees (○). The size of larger trees (diameter > 15 cm) is indicated by symbol sizes proportional to the DBH (excluding stumps, which are proportional to the basal diameters). Smaller trees are show with symbols corresponding to a diameter of 15 cm. Panel (D) shows the (0–1 normalized) intensity of large pine trees in the 2006 pre-fire stand (estimated using an Epanechnikow kernel with bandwidth 10 m; Stoyan and Stoyan, 1994) together with the pre-fire spatial pattern of all small pine (white) and birch (black) trees.

$g_{12}(r)$ to quantify the observed spatial patterns (Stoyan and Stoyan, 1994; Condit et al., 2000; Wiegand and Moloney, 2004; Illian et al., 2008). For homogeneous patterns, the univariate pair correlation function $g(r)$ can be defined as the expected density of points in a ring of radius r and width dw centered in an arbitrary point, divided by intensity λ of the pattern (Stoyan and Stoyan, 1994). Values of $g(r)$ that are more, less than, or equal to 1 indicate spatial aggregation, regularity, or randomness, respectively. The bivariate $g_{12}(r)$ describes the spatial relationship between two patterns (pattern 1 and pattern 2). The $g_{12}(r)$ can be defined as the expected density of type 2 points in a ring of radius r and width dw centred in an arbitrary type 1 point, divided by the intensity λ_2 of the pattern of type 2 points (Wiegand and Moloney, 2004). With this terminology, g_{22} refers to the univariate pair correlation function of type 2 points.

All analyses were carried out using the grid-based software Programita for point spatial pattern analysis (Wiegand and Moloney, 2004; Wiegand et al., 2007). We used for all analyses a cell size of 0.5 m × 0.5 m. This spatial resolution was fine enough to respond to our objectives. For estimation of the pair-correlation functions we used a ring width of 1 m. Details on the estimators of the pair-correlation function and edge correction can be found in Wiegand and Moloney (2004).

To assess departures from the null model we compared the pair correlation functions of the observed spatial patterns with approximately 5% simulation envelopes being the fifth lowest and highest values of the pair correlation functions of data created by 199 simulations of appropriate null models (described below). Note that the simulation envelopes cannot be interpreted as confidence intervals for formal hypothesis testing because type I error inflation may occur due to simultaneous inference (i.e., tests at many spatial scales; Diggle, 2003). However, this is of minor concern for our exploratory data analysis.

2.4. Univariate analyses (analysis 2)

We compared the univariate spatial patterns to the null models of complete spatial randomness (CSR) or to a Thomas processes that describe clustering (Thomas, 1949; Wiegand et al., 2007). In the 2006-burned stand (Fig. 1A) and the unburned stand (Fig. 1C) a cluster process was chosen because the pattern of small trees showed apparently strong spatial clustering. The Thomas process is able to reveal additional characteristics of the clustering (Wiegand et al., 2007). Details on the Thomas processes can be found at appendix A and Wiegand et al. (2007). To test if the Thomas process provided a reasonable fit we used the accumulative distribution function of the distances y to the nearest neighbor $G(y)$ as additional test statistic as e.g., done in Wiegand et al. (2007).

2.5. Bivariate analyses (analysis 3)

To investigate the relationship between small and large trees (and trees of different species for the 2006-burned stand) we used the toroidal shift null model (Goreaud and Pelissier, 2003; Wiegand and Moloney, 2004) which provides a non-parametric way to test for independence between two patterns. This null model treats the study plot as torus and involves random shifts of the whole of pattern 2 relative to pattern 1 which is kept fixed (Dixon, 2002). If the observed bivariate pair-correlation function $g_{12}(r)$ was above or below the simulation envelopes the two patterns showed attraction or repulsion, respectively.

2.6. Test of random mortality hypothesis (analysis 4)

To explore possible non-random spatial structures in tree mortality we used random labeling as null model (Kenkel, 1988; Goreaud and Pelissier, 2003; Illian et al., 2008). This null model assumes that mortality acted as a random process over a given tree pattern, i.e., the n_2 dead trees of a stand are assumed to be a random subset of the joined pattern of the n_2 dead and n_1 surviving trees (1 referring to surviving and 2 to dead trees, respectively). If, e.g., death rates among neighboring trees are elevated due to competition (or elevated ignition probabilities) we would expect that the pattern of dead trees should be more clumped than the pre-mortality pattern (e.g., Kenkel, 1988; Getzin et al., 2006). The test of the random mortality hypothesis was conducted by computing appropriate test statistics (see below) from the observed data, then randomly re-sampling sets of n_2 trees from the joined pattern of dead and surviving trees to generate simulation envelopes of the test statistic.

We used three test statistics to test for possible departures from the random mortality hypothesis. The first test statistics $g_{22}(r)$ is the univariate pair correlation function of the pattern of dead trees (the subscript 2 refers to dead trees) and tests if the pattern of dead trees shows a spatial structure, additionally to that of the pre-mortality trees. Note that the expectation of this null model is given by the univariate pair correlation function of the pattern of pre-mortality trees [i.e., $g_{1+2,1+2}(r)$ where 1 + 2 symbolizes the joined pattern of dead and surviving trees]. If the $g_{22}(r)$ was above the simulation envelopes dead trees showed additional clustering, whereas dead trees are more regularly than the joined pattern if $g_{22}(r)$ was below the simulation envelope. The second test statistic $g_{12}(r)$ is tailored to detect correlation between dead (type 2) and surviving (type 1) trees (Goreaud and Pelissier, 2003). The test statistic $g_{12}(r)$ is below the simulation envelopes if there are less dead neighbors at distance r of an arbitrary surviving tree than expected under random labeling. This indicates that surviving and dead trees tend to be negatively correlated at distance r (i.e., segregation). Conversely, surviving and dead trees are positively correlated if the test statistic $g_{12}(r)$ is above the simulation envelopes.

We developed a new test statistics which specifically tests for density-dependent effects in mortality. The difference $g_{2,1+2} - g_{1,1+2}$ compares the density of pre-mortality trees around dead trees (pattern 2) with the corresponding density of pre-mortality trees around surviving trees (pattern 1). The expected value of this test statistics is zero under random labeling, but under (negative) density-dependent mortality we expect that dead trees would occur preferably in areas with high pre-mortality tree densities, i.e., $g_{2,1+2} - g_{1,1+2} > 0$.

3. Results

3.1. Surface fire characteristics

For larger trees (i.e., the classes DBH >10 cm, >20 cm, >30 cm, and >40 cm) mean bole char height was in the 2006-burned stand significantly larger than in the 1994-burned stand (Table 1). However, the one-way Kruskal–Wallis test showed that mean bole char heights of the two burned stands were not significantly different when using all charring stems (i.e., small and large trees; $p = 0.0152$). As expected, mean bole char heights had a tendency to increase with size classes; the correlation between mean char heights and size class was significant in the 2006-burned stand (Spearman rank coefficient $R = 0.657$, $p < 0.0001$) and positively insignificant in the 1994-burned stand (Spearman rank correlation coefficient $R = 0.740$, $p = 0.2600$).

3.2. Stand structure

The forests in our three study plots were characterized by a common appearance and similar species compositions (Tables 1 and 2). All three stands showed similar mean DBH values for size classes with a DBH larger than 10 cm, but stark differences in the abundance of stems less than 10 cm in DBH (Table 1). In the 2006-burned stand, 88% of small Mongolian pines (<15 cm DBH) died during the surface fire, but none of the large pines died (Table 3). Interestingly, the number of large surviving pine trees was approximately the same in all three stands (Table 3). Density of surviving trees was almost one fourth of the pre-mortality rivals in the 2006-burned stand, while nearly 3% of basal areas were killed.

Table 1
Kruskal–Wallis test and Duncan's multiple range tests to compare mean DBH and mean bole char heights for different size classes among the three stands. The standard errors are given in parenthesis. The same letter indicates that the means of a column have no significant difference at the 0.05 level. Standing dead trees and fallen logs with chars were included in the calculation of average char heights in the 1994-burned stand.

Plot	All size classes		DBH ≥ 10 cm		DBH ≥ 20 cm		DBH ≥ 30 cm		DBH ≥ 40 cm	
	N	Mean	*N*	Mean	*N*	Mean	*N*	Mean	*N*	Mean
DBH										
Pre-2006-burned	2254	7.1(0.24)	363	30.6(0.55)[a]	304	34.1(0.43)[a]	215	37.4(0.42)[a, b]	56	46.0(0.70)[a]
Post-2006-burned	567	21.6(0.62)	360	30.8(0.54)[a]	304	34.1(0.43)[a]	215	37.4(0.42)[a, b]	56	46.0(0.70)[a]
1994-burned	379	28.1(0.57)	341	30.6(0.47)[a]	295	32.9(0.39)[a]	190	37.0(0.32)[a]	46	42.8(0.38)[b]
Fire-excluded	871	15.3(0.42)	479	23.6(0.51)[b]	239	33.0(0.54)[a]	147	38.6(0.42)[b]	59	43.6(0.41)[b]
Char height										
2006-burned	2253	150.1(2.68)[a]	363	343.7(7.85)[a]	304	364.6(8.59)[a]	215	374.8(10.36)[a]	56	391.5(21.95)[a]
1994-burned	481	139.7(3.50)[a]	384	152.6(3.88)[b]	304	164.1(4.35)[b]	192	169.1(5.58)[b]	46	164.4(12.61)[b]

Table 2
Species composition, tree density, and basal area for the three stands. The percentages are given in parenthesis. For the stand burned in 2006, values are given before and after the fire.

Plot	*Armeniaca sibirica*	*Crataegus dahurica*	*Malus baccata*	*Betula platyphylla*	*Pinus sylvestris* var. mongolica	All species
Density (trees ha^{-1})						
Pre-2006-burned		1(0.0444)	6(0.2662)	624(27.6841)	1623(72.0053)	2254
Post-2006-burned		1(0.18)		138(24.34)	428(75.48)	567
1994-burned				1(0.27)	378(99.73)	379
Fire-excluded	1(0.11)		16(1.84)	2(0.23)	852(97.82)	871
Basal area (m^2 ha^{-1})						
Pre-2006-burned		0.0002(0.0006)	0.0006(0.0019)	0.8424(2.6898)	30.4748(97.3077)	31.318
Post-2006-burned		0.0002(0.0006)		0.4386(1.4432)	29.953(98.5562)	30.3918
1994-burned				0.0324(0.12)	27.1758(99.88)	27.2082
Fire-excluded	0.0001(0.0004)		0.0206(0.077)	0.0001(0.0004)	26.7221(99.9222)	26.7429

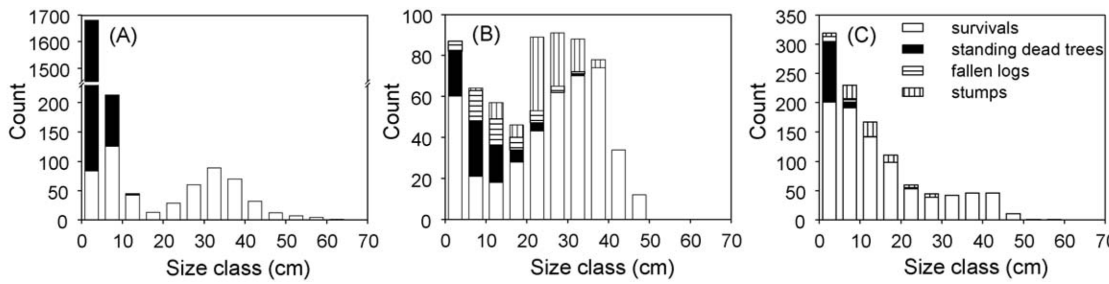

Fig. 2. Size class distributions and tree conditions for the three stands. (A) the 2006-burned stand, (B) the 1994-burned stand, and (C) the fire-excluded stand. Symbols for surviving trees (open bars), standing dead trees (closed bar), stumps (vertical stripe), and fallen logs (horizontal stripe). Note that conversion of diameter at the base data to DBH was carried out in some cases for stumps.

In the 2006-fire event, the values of Mongolian pine and birch fell by 73.6% and 77.9% in density and by 1.7% and 47.9% in basal area, respectively. Among our three plots, the fire-excluded stand had minimal basal area and maximal amount of survivals, and the 1994-burned stand consisted of nearly pure Mongolian pines (only one birch left) with the least woody species, but had the least stems of the three stands studied. It is interesting to note that saplings at the 1994-burned stand (i.e., trees <1.3 m in height), which were almost all of Mongolian pines and older than 2 years, reached surprising densities of some 17,476 trees ha^{-1}, but that the number of small pine trees was far the lowest among the three stands (Table 3).

In the 2006-burned stand, the number of dead trees was about three times higher than the number of survivors; however, all dead trees were smaller than 16 cm in DBH, and 89% of the dead stems had a DBH ≤4 cm (Table 1, Fig. 2A). In the stand that burned in 1994 we found that 37% percent of all stems died gradually, and the replaced stems were mainly smaller size class of DBH ≤20 cm (56.5% of all replaced stems) (Fig. 2B).

3.3. Univariate point pattern analysis

Large pines showed in all three stands a tendency of regularity at small scales, both for the pre- and the post-mortality patterns (Fig. 3B, D and F) confirming hypothesis 2. However, the univariate analysis did not detect stark differences in the spatial patterns for large pines with time since fire (the 2006 pattern showed a non-significant tendency to regularity and the two other plots weakly significant regularity). Surprisingly, the intensity of surviving large pines was very similar among the three plot (316, 323, and 338 stems ha^{-1} at the 2006, 1994, and unburned plot, respectively). The 2006 burned and the unburned plot showed very low mortality of large pines (0% and 7%, respectively) whereas 24% of the large pines died at the 1994 burned plot, pointing to stronger density-dependent mortality in this stand.

Table 3
Number of small (DBH <15 cm) and large (DBH ≥15 cm) stems of Mongolian pine at the three sites.

Plot	Small pine		Large pine		Small pine	Large pine	Pine
	Surviving	Dead	Surviving	Dead	All	All	All
Fire-excluded	514	183	338	25	697	363	1060
1994-burned	56	109	323	113	165	436	601
2006-burned	115	1195	313	0	1310	313	1623

In contrast to large trees, we found marked differences in the spatial pattern of small trees with respect to time since fire. Small pines showed at the 2006 pre-fire plot and the unburned plot strongly clustered patterns (Fig. 3A and E). However, at the 1994 burned stand, where small pines had a low intensity, we found random spatial patterns (Fig. 3C). The Thomas process provided a good overall fit of the spatial patterns of small pines in the 2006 stand, but the pair-correlation function (Fig. 3A) and the distribution function of the distances to the nearest neighbor (Fig. A1) indicated an additional small-scale clustering at scales smaller than 2 m in the pre-fire pattern. Additionally, a few surviving small pines (post-fire pattern) had their nearest neighbor further away than expected by the fitted Thomas process (Fig. A1). Fit with the Thomas cluster process to the pre-fire data of the 2006 stand revealed a cluster radius of 10.6 m and 16 clusters. Fit with the Thomas cluster process to the post-fire data of the 2006 stand revealed a cluster radius of 11.6 m but only and 4 clusters. Thus, the cluster size did not change but the trees died in clusters.

Similarly, the small pine trees of the unburned stand could be described well with a Thomas cluster process with cluster radius of 7 m and 140 clusters, but there was also some additional small-scale clustering at scales <2 m (Fig. 3E; Fig. A1). At this stand, however, the post-mortality pattern is somewhat less clustered than the pre-mortality pattern (Fig. 3E).

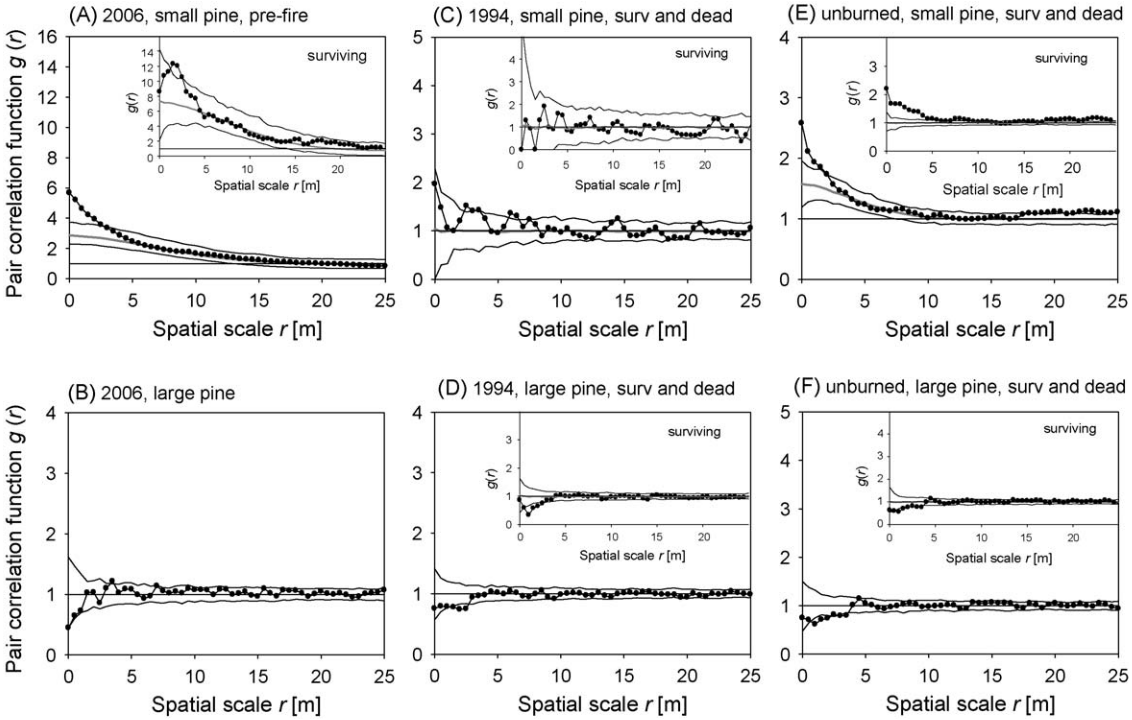

Fig. 3. Univariate point pattern analyses. The pair-correlation function of the data is shown as solid line with closed circles, the expectation of the null model as solid grey line, and the simulation envelopes being the fifth smallest and largest values of the 199 simulation of the null model are shown as solid lines. The main figure shows the analysis of the joined pattern of surviving and dead trees and the small insets show the corresponding analysis of the surviving trees. In (A) and (E) we used a Thomas process as null model and in all other analysis complete spatial randomness (CSR).

3.4. Bivariate point pattern analysis

As expected, we found significant small-scale repulsion between the large number of small pines and large pines at the pre-fire 2006 stand, which however disappeared after the fire (Fig. 4A). Fig. 1D shows that small pines occur in areas of low intensity of large pine trees, pointing to strong competitive interactions. However, contrary to our expectation we found that small and large pines were independent at the other two stands (Fig. 4B and C) although with a tendency to attraction at the unburned stand (Fig. 4C).

3.5. Birch patterns

Birch, which occurred only at the 2006 stand, behaved quite similar to small pines. It showed a particular spatial pattern with cluster radii of approximately 13–17 m, but very strong small-scale clustering at scales smaller than 2 m which only partly disappeared after the fire (Fig. 5A; Fig. A1). Interestingly, the pre-fire birch and small pine patterns showed small-scale repulsion, which disappeared after the fire (Fig. 5B). Similarly to small pine, birch showed also a significant small-scale repulsion to large pine trees (Fig. 5C). The influence of large pine trees on the pattern of birch and small pines is illustrated in Fig. 1D.

3.6. Random mortality hypothesis

Dead small pines at the 2006 stand were at scales <2 m surrounded by more pre-mortality small pines than surviving small pines (Fig. 6A). Surviving small pines at the 2006 stand were strongly aggregated at small scales (Fig. 6B), and surviving and dead small pines were spatially segregated (Fig. 6C). The

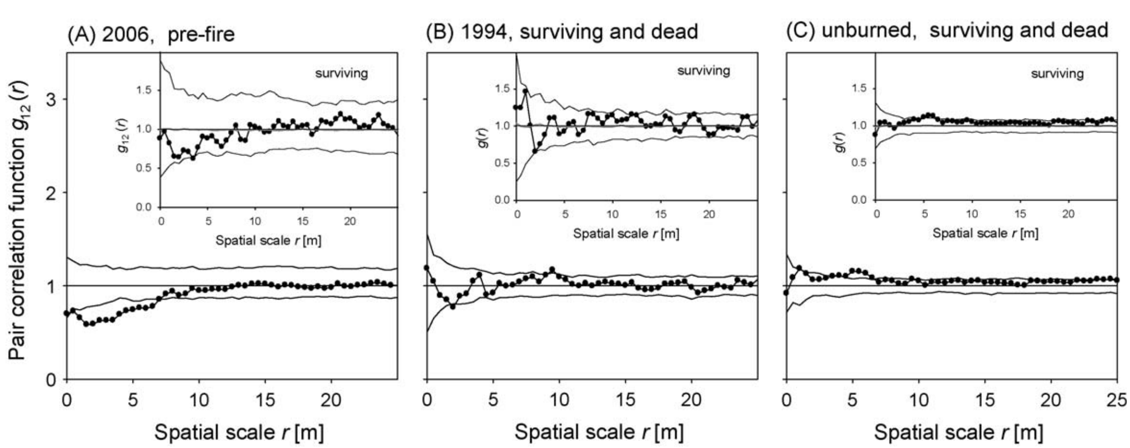

Fig. 4. Bivariate analysis showed the association between small and large pine trees. We used the torodial shift null model to test for independence. Other conventions as in Fig. 3.

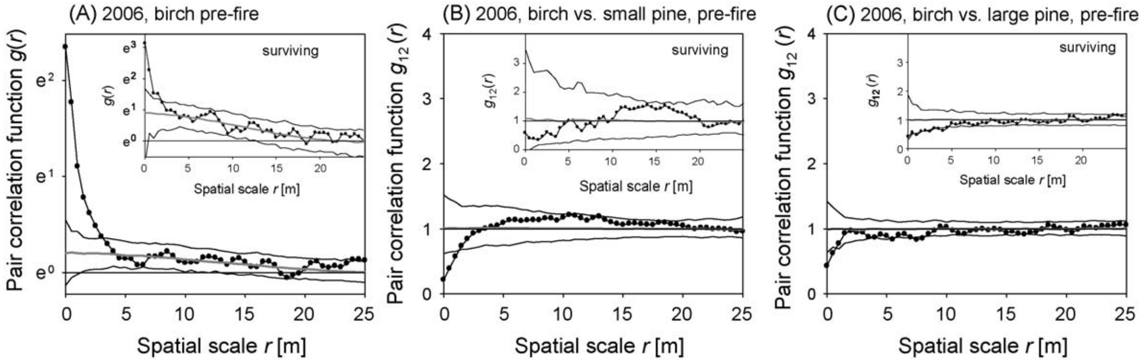

Fig. 5. Analysis of birch in the 2006 stand. (A) Univariate analysis using a Thomas process as null model. (B) Spatial association between birch and small pine using the torodial shift null model testing for independence. (C) Spatial association between birch and large pine using the torodial shift null model testing for independence. Other conventions as in Fig. 3.

segregation of surviving and dead small pines points to favorable micro-sites for survival. Note that these results are opposed to the expectation that the post-fire pattern is more regular than the pre-fire pattern. The birch at the 2006 stand showed a behavior similar to small pines. The surviving birch trees were aggregated (Fig. 6E), surviving and dead birch trees were spatially segregated (Fig. 6F), and there was a weakly significant effect of neighborhood birch density on mortality (Fig. 6D).

At the 1994 stand departures from random mortality occurred for large pines (Fig. 6G–I) but not for small pines (Fig. 6J–L). Dead large pines were significantly aggregated within all surviving and dead large pines at scales <5 m (Fig. 6H) as expected under self-thinning (e.g., Kenkel, 1988). This result was strengthened by the analysis with the test statistics $g_{2,1+2} - g_{1,1+2}$ which revealed density-dependent effects at small scales where dead large pine trees had more pre-mortality neighbors than surviving ones (Fig. 6G).

At the unburned stand we found aggregation of dead small pines (Fig. 6N) and a clear density-dependent mortality of dead small pines that were located in areas of higher density of small pre-mortality pines (Fig. 6M). Interestingly, surviving and dead small trees were not spatially segregated (Fig. 6O) as observed at the 2006 burned stand (Fig. 6C and F).

4. Discussion

Fire is a prevailing disturbance in Mongolian pine forests (Zhao and Li, 1963); however, there is little information available on the variations of stand structure and spatial pattern in this type of forest. To close this gap we performed a detailed analysis of stand structure characteristics, spatial patterns, and spatial effects in density-dependent mortality based on data of three 1 ha plots of a Mongolian pine forests in the Hulun Buir sandland.

4.1. Variability in stand structure

The three pine forest stands that regenerated from severe disturbances since the 1950s (Zhao and Li, 1963) showed similar upperstory structures, entirely dominated by Mongolian pine, and similar size structure (e.g., mean DBH ≥20 cm of three stands showed no significant difference, Tables 1 and 3). Surface fire killed at our stands basically trees of the lower stratum (Bond and van Wilgen, 1996; Zenner, 2005) and smaller upperstorey trees, thereby reducing the quantity of understory woody species and smaller stems dramatically.

Our analysis supported our first hypothesis, i.e., that stand structure characteristics would change substantially with time since fire. Tree densities were decreasing at the 2006-burned stand almost threefold (McNicoll and Hann, 2003), and the diameter distributions shifted from a rotated sigmoid shape to a typical bimodal shape (Zenner, 2005) with less positive skew. At the 1994-burned stand the drop in post-fire tree density resulted in a diameter distribution with less negative skew. Finally, the diameter distribution recovered positive skew in the fire-excluded stand (Tables 1 and 2, Fig. 2). However, mean DBH changed only slightly with respect to time since fire and remained relatively stable (Tables 1 and 2). Surface fire may thus be a "pair of scissors" to cut dense smaller trees (Fulé and Laughlin, 2007) and to open the lower layer of the given stand to trigger the next generations of canopy stems. Our study forests are thus held by surface fire on an appropriate density, pushing it on the long-term in composition and structure to that of a mature forest ecosystem. Contrary to other studies (e.g., Sibold et al., 2007), the trajectory of stand density across our forest presents strong evidence that stand structure changed significantly over time and exhibited strong self-thinning without fire.

4.2. Spatial patterns and self-thinning

Many studies have shown that density-dependent mortality may cause stem distribution to shift from an aggregated pattern to random, and then to repulsion (Daniels, 1978; Kenkel, 1988; Mast and Veblen, 1999; He and Duncan, 2000; Kashian et al., 2005). The trend to regularity in the spatial patterns of large pine supports the prevailing view of regular mature forests (Pielou, 1961; Sterner et al., 1986; Duncan, 1991; Shackleton, 2002; Stoll and Bergius, 2005), however, we could not detect the expected increase in regularity with time since fire (hypothesis 2). Analyses of the overstocked 2006 stand pointed to a different mechanism of self-thinning. Fire mortality did not change the typical cluster size of the pattern of small pines, but reduced the number of clusters (analysis 2). Analysis 4 confirmed this diagnosis, showing that surviving small pines were clustered and segregated from dead small pines (Fig. 6A–C). Thus, surviving small pine trees were clustered in a few favorable micro-sites that provided protection from fire. This suggests that the effect of fire on mortality of small stems was spatially heterogeneous, leaving some of them alive thereby guaranteeing regeneration.

Interestingly, basal area of Mongolian pine, which is a surrogate for total timber mass, decreased only slightly with time since fire (i.e., from pre-2006, post-2006, 1994-burned, to unburned; Table 2) and the number of surviving large pines was very similar among the three plot (Table 3). One reason for this might be that fire did only rarely kill large pines and that all three stands had approximately the same age. Similarly, Fulé and Laughlin (2007) observed low declines in basal area after fire for ponderosa pine

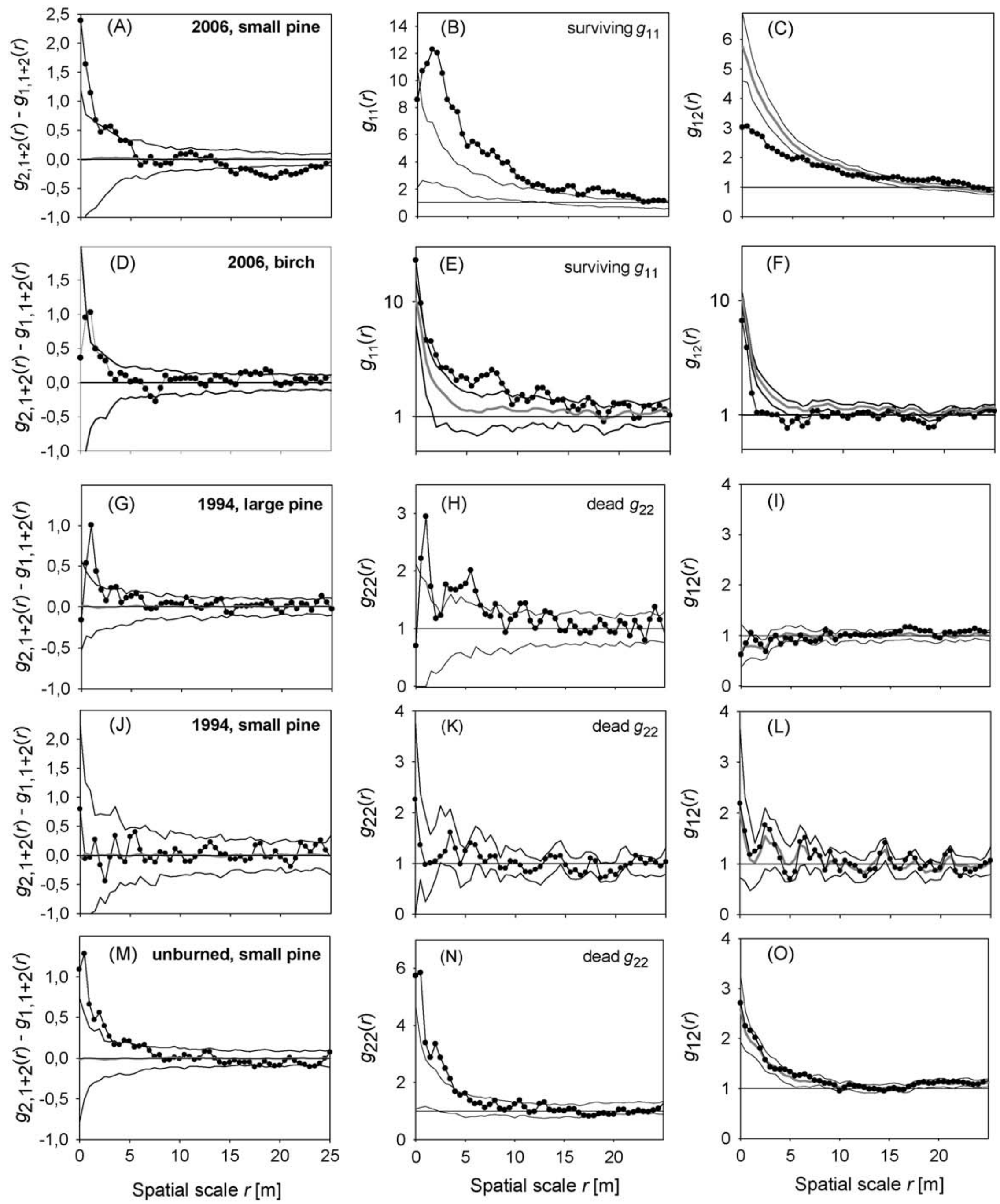

Fig. 6. Analysis of random mortality. The different columns show different test statistics to detect departures from random mortality and each line shows a different pattern for which we aim to find departures. Left column: test statistics to reveal density dependence in mortality. If the test statistic showed positive values if tree density around dead trees is larger than expected under random mortality. Middle column: test statistic to reveal aggregation of dead or surviving trees. If $g_{22}(r)$ is above (or below) the simulation envelopes then dead (or surviving) trees are more aggregated than expected under random mortality. Right column: test statistic to detect spatial segregation between surviving and dead trees. If $g_{12}(r)$ is below the simulation envelopes, dead and surviving trees are spatially segregated. Other conventions as in Fig. 3.

(*Pinus ponderosa*) and Douglas fir (*Pseudotsuga menziesii*), but considerable thinning effect of fire on smaller, shorter, fire-susceptible trees. However, although large pines did not show significantly regular patterns they showed clearly less aggregated spatial patterns than small pines.

In contrast to large pines, the spatial patterns of small trees exhibited with respect to time since fire changes from complex patterns with two critical scales of clustering to random, and then again to clustered patterns (Fig. 3A, C and E). The lack of clustering in small pines of the 1994 stand might be explained by the unusually high densities of saplings (17,476 trees ha^{-1}) which were not included in our analysis. Only the overstocked 2006 pre-fire stand showed clear evidence for the expected negative spatial association between small and large trees (hypothesis 3). For example, Montes and Cañellas (2007) found that the sapling pattern of Scots pine (*P. sylvestris* L.), a generic of our study species, showed repulsion to larger trees which was due to suppression of sapling development near the mother trees. However, this repulsion disappeared after the 2006 fire. Thus, fire selectively eliminated dense clusters of small trees in areas of low density of large trees.

Analysis of the spatial pattern of mortality using random labeling clearly rejected the random mortality hypothesis (Kenkel,

1988) for all size classes and stands (hypothesis 4), except the small pine population at the 1994 burned stand (Fig. 6J–L). To study the particular pattern of dead trees in more detail we used a novel test statistic for the random labeling null model which was especially designed to detect density-dependent effects in mortality. This test statistic proved to be a valuable complement to common test statistics e.g., introduced by Kenkel (1988) and discussed in Goreaud and Pelissier (2003) and very efficient in detecting density-dependent effects.

We found very clear evidence for density-dependent effects in mortality: small trees with more neighbors were more likely to die. This was true for fire-induced mortality at the 2006 burned stand, both for pine and birch, and for the two stands that were fire-free for some time. In case of surface fire, fire may propagate better in areas of high stem density where more fuel is available (Fulé and Covington, 1998; Rebertus et al., 1989). This suggests a strong fire-induced thinning at the stand scale (Cissel et al., 1999) which speeds up the creation of mature forest structures. In the two other stands where density of small pines was lower, competition may be more intense in areas of higher stem density thereby causing the observed pattern. This was true for small pines at the unburned stand and for large pines in the 1994 stand (which had higher pre-mortality densities of large pines than the two other stands). Our analyses thus confirmed our fourth hypothesis (i.e., mortality should be spatially correlated in a way that stems located in areas of higher stem density have a higher likelihood of mortality).

The 1994 stand is at a phase where density-dependent mortality occurred for large pines, increasing its degree of regularity, but where mortality was random for small pines which showed also random spatial patterns. In contrast, the 2006 pre-fire and the unburned stand were structurally similar. Both showed aggregated patterns of small trees, regular pattern of large pine, and density-dependent mortality of small trees. At the 1994 burned stand a large sapling population will probably within a few years enter the small tree size class and show similar patterns as in the 2006 pre-fire stand. As demonstrated by the 2006 burned stand, fire kills most of the "overstocked" of small trees in a density-dependent way thereby removing the pre-fire repulsion between small (pine or birch) and large trees, converting it into independent post-fire patterns.

4.3. Management implications

The observed impact of surface fires on Mongolian pine forests of the Hulun Buir sandland allows assessing the ecological consequences of forest management and continued fire suppression practices. Surface fire plays a key role in interrupting succession towards overstocked forests by reducing stand density and speeding up succession towards a mature forest. These observations motivate a management approach that permits moderate harvesting the smaller trees (i.e., poles) in overstocked forest stands (Tappeiner et al., 1997) for fuel purposes in rural areas (Weyerhaeuser et al., 2005). This would be especially relevant in countries such as China with severe shortages of timber and fuelwood (Yao, 2000).

Allowing harvest of poles may also have important implications for nationwide forest restoration under the current forestry policy, i.e., the six state key ecological programs initiated since 1998 in China, which mainly focus on ecological restoration (Zhang et al., 2000) and strictly ban commercial logging (Xu et al., 2006). Therefore, in contrast to the prevalent philosophy and practice in the region studied, it would not be necessary to extirpate surface fire with great labor and costs. On the other hand, fire suppression in coniferous forests systems may lead to accumulation of fuel and to large catastrophic fires that cannot be effectively controlled (Sturtevant et al., 2004). Thus, moderate extraction of poles may reduce this risk and avoid costly fire extirpation programs.

In summary, our study on Mongolian pine forest stands highlights the importance of surface-fire-induced thinning as a way to alleviate direct competitive interaction of individuals and to create open space for survivals to dwell in, as shown by the high sapling density at the 1994 burned stand. As a result, surface-fire-mediated forests will likely benefit from the fire disturbances and help the stands not only to escape from an expected increases in fire size and severity due to changing climate, but also to direct them to the mature or old-growth structural pathways.

Acknowledgements

We are grateful to those who helped us to map countless numbers of live and dead trees and conduct other field tasks: Chuntang Yu, Lei Mao, and Wenyi Wu. The field survey was also partially supported by the Honghuaerji Forestry Administration Bureau. Partial funding was provided by the National Natural Science Fund (30571529) and State Sci. &Tech. Support Project (2006BAD26B05). We also thank two anonymous reviewers for their helpful comments.

Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at doi:10.1016/j.foreco.2009.02.019.

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